

Surprising facts about fascial physiology and biochemistry

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Introduction

Myofascial tissue is composed of cellular and non-cellular elements. Both components are sensitive to physical strain, temperature, pH and humoral factors. This chapter describes the physiological and biochemical properties of fascial tissues, as well as the interaction with connected tissues, most notably skeletal muscle, and repair mechanisms after injury. Knowledge of underlying neurophysiological pathways is indispensable because “you can’t depend on your eyes when your imagination is out of focus” (Mark Twain, 1889).

Neurophysiological fundamentals

Movement in sport, first and foremost, involves a sophisticated interaction of the central and peripheral central nervous system, force generating muscle and connective tissues, such as fascia, which is intrinsically tied to skeletal muscle.

Upper and lower motoneuron

The protocol of a movement (engram) is generated in the motor cortex of the brain. The electrical impulses originating from pyramidal neurons propagate along their axons, crossing over to the contralateral side on the brainstem level. This is the reason why the left hemisphere of the brain controls the right side of the body. In most cases, the left hemisphere is dominant. This is evident not only in right-handed people

but, surprisingly, also in the vast majority of left-handed individuals. Aside from the brain, left predominance is found in other matched organs such as kidneys or testicles, which are slightly bigger on the left side. The heart is also localized on the left. The asymmetry of our bodies originates from the directional movements of cellular fimbriae in an early stage of fetal development.

Nerve fibers of the pyramidal tract, also denominated corticospinal tract or upper motoneurons, carry information from the brain to the spinal cord. The electrical signal from the upper motoneuron is then transmitted in the anterior horn of the spinal cord to the lower motoneuron, which finally carries the information to the skeletal muscle (Figure 2.1). By definition, an injury of the upper motoneuron leads to a spastic palsy due to disinhibition of motor reflexes. By contrast, lower motoneuron lesion results in flaccid paralysis. Connective tissue reacts to such disturbances of the reflex circuit in due course. For example, after damage to upper motoneurons, as in stroke or spinal cord injury, there is significant fascial stiffening, which in some cases may require surgical relief of the strain. The myofascial stiffening is the result of hyperreflexia due to the lesion of the central nervous system, most notably the upper motoneurons. In summary, two nerve cells, including their long axons, carry the motor information from the brain to the muscle. On the sensory side, three nerve cells are assembled. The peripheral sensory neuron transmits the information to a second neuron in the

Chapter 2

dorsal horn of the spinal cord. Here, the signal is forwarded, through the spinal cord, to a nut-like part of the brain called the thalamic nuclei. This is the central part of the sensory system, which is composed of two interconnected neurons. The thalamus is located at the center of the brain and is also known as “the door to consciousness”.

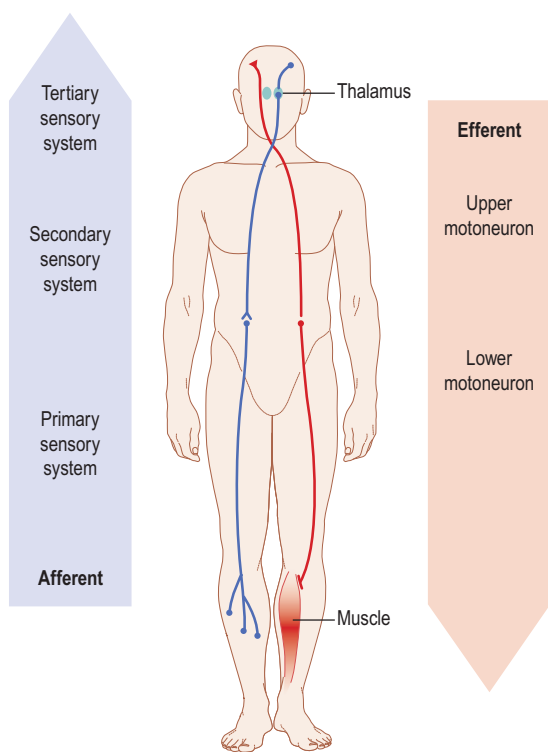


Figure 2.1 Basic neuronal elements of movement control

Two efferent nerve cells, upper and lower motor neuron, conduct specific movement information (encoded as action potential frequency) to the target organ, i.e. skeletal muscle. On the sensory side, there are three neurons interconnected. The extrapyramidal system integrates information such as optic, acoustic and sensory input and modulates the efferent corticospinal tract, i.e. the upper motoneuron. A major component of the extrapyramidal motor system is the thalamic nuclei. Formation of peripheral nerve plexus is not shown in the figure.

Extrapyramidal movement coordination

This system of motor control is subject to a variety of influences and modifications, which are collectively known as the extrapyramidal motor system. Important anatomical components of the extrapyramidal motor system are the basal ganglia, the cerebellum and numerous interneurons, which interconnect pyramidal axons recursively, side by side and with brain nuclei. The extrapyramidal system is designed to enhance or hamper signal transduction. It receives signals from the somatosensory input, and coordinates and fine-tunes movements (Chaitow & DeLany, 2000). A loss of function of this system, e.g. in Parkinson's disease, leads to a very specific akinetic and rigid movement disorder.

Interestingly, other inputs converging in the thalamic nuclei also influence the extrapyramidal movement coordination. Here, important factors are emotion and pain. Emotion is believed to be generated by a circular flow of electrical impulses along a set of anatomical brain structures known as the limbic system. Furthermore, the limbic system is a key element for memory, olfaction and evolutionary survival. It includes several anatomical brain structures, notably interfering with the thalamic nuclei and thereby giving an explanation as to why emotional state can influence movements and vice versa. In this context, it is interesting that regular sporting activity improves learning, reduces the probability of dementia, and significantly prolongs life expectancy.

Cerebral hormones such as dopamine and serotonin are key promoters for movement but also for happiness. For example, this coherence can be observed in patients with psychiatric mania, who are restlessly in motion until they are treated with neuroleptic drugs, i.e. dopamine antagonists, after which rather robotic movements may

Surprising facts about fascial physiology and biochemistry

be observed. On the positive side, good mood is often associated with physical activity, such as dancing. Another example is the need for movement in children, especially as an expression of happiness. The direct effects of these hormones on fascial tissue have only been shown indirectly to date. For instance, long-term medication with potent dopamine agonists leads to fibrosis in the lungs and around the kidneys, as well as thickening and stiffening of cardiac valves.

Pain is a complex topic involving humoral factors, peripheral nervous structures and higher brain centers, e.g. thalamic nuclei. Fascia contains numerous nociceptors such as free nerve endings, whose stimulation triggers pain sensations. These nerve endings are linked to the autonomic nervous system (ANS), which comprises the sympathetic and the parasympathetic systems. ANS collects the information from the body and the external environment, and activates specific responses in the body.

In contrast to the aforementioned somatosensory neurons, which have an impulse conduction speed of 10–100 m/s, a specific type of nociceptor, the C-fibers, are not surrounded by a myelin sheet and, therefore, conduct impulses more slowly, at roughly 0.5 to 2 m/s. The resulting pain sensation is dull and less localized. Fascial tissue mostly contains unmyelinated nerves with free nerve endings serving as unspecific receptors. These free nerve endings detect several stimuli such as tension, temperature or pain, and are connected to wide dynamic range neurons for further processing of the input.

Irritation and coupling of sympathetic fibers with sensory fibers is one of the known mechanisms for the development of chronic pain, such as in the Complex Regional Pain Syndrome (also called Sudeck's disease). Moreover, fascia also contains neurons that release humoral mediators,

such as substance P or calcitonin-gene related peptide, which are associated with chronic and self-sustaining pain (Tesarz et al., 2011). Further details can be found in Chapter 15.

Muscle and fascia – together strong

Skeletal muscle is composed of myofibers, which develop as a merger of several single muscle cells via multiple cell fusions. Microscopically, myofibers are readily identifiable by the regular arrangement of the contractile proteins. The subcellular contractile units are called sarcomeres, in which effective force generation requires optimum interdigitation of actin and myosin filaments. *In vitro* experiments have shown that, after surgical excision of a muscle bundle, the elastic fibers shrink by roughly one third of their length *in vivo*. Therefore, *in vitro* force measurements in a laboratory cell-dish environment require an experimental pre-stretching of the bundle.

For efficient contraction development, this pre-stretch may be dynamically regulated in the body by fascial tissue. Depending on the cross-link tension, hydration, elastic content, and even active fascial contraction, the strain on myofibers can be regulated in order to intensify or weaken muscular force generation. In other words, fascia acts as an energy-saving and intelligent servomechanism.

Myofascial tone

Muscle tone is a feature that is regularly assessed by a broad variety of medical professionals. However, diagnosis is based on the experience of the examiner, as reference standards are lacking for symptoms such as body aches or subjective stiffness. Manually assessed excessive muscle tone is also often attributed to postural pain syndromes. Systematically, it is more precise

Chapter 2

to use the term “myofascial tone” because the tensional properties originate not only from the muscle tissue itself but also from the fascial tissue (i.e. the epi-, peri- and endomysium), which can alter its mechanical properties (see below). Interestingly, the amount of connective tissue found in a specific muscle is dependent on its function. Tonic muscle contains significantly more and firmer fascial components than phasic muscle (Klingler et al., 2012). Differences are also found between species. For example, meat from mountain goats has a much higher percentage of collagenous connective tissue compared with domestic pigs. In hereditary muscle diseases, fibrosis is one of the most prominent features and may even be the first symptom of the disease. Therapeutic efforts aim to reduce the fibrosis in muscle disease, such as Duchenne myopathy. Here, corticosteroids are effective inhibitors of fibroblast proliferation and collagen synthesis. Corticosteroid treatment then significantly postpones the onset of disability, in relation to walking. Due to the adverse effects of high corticosteroid doses, alternative treatment options are currently being explored for the management of such pathologies (Klingler et al., 2012). In summary, the contribution of fascia and myofibers to muscle tone is highly variable and adaptive to external and internal factors.

Resting tone

What is resting myofascial tone? This question is not trivial and will be addressed in the following paragraph. By definition, the resting skeletal muscle is electrically silent. This means that the voluntary innervation is zero and the reflex circuit is quiescent. In theory, this is a clear condition and should be easy to assess by examining a resting muscle in terms of resistance to passive movement and firmness of the tissue (Masi and Hannon, 2008). However, from a practical point

of view, it can be observed that in awake and even sleeping subjects resting muscles appear to be far from silent. In most patients who are examined by means of electromyography (EMG), signs of neural activity in the muscle tissue can be seen, even if the muscle is voluntarily slack. Indeed, the creaky sound of the EMG device can be used as a biofeedback method in order to calm and relax the patient. Another problem during clinical examination is that many people tend to innervate (“unconscious helping”) the muscle/limb under investigation or suffer from disinhibition of monosynaptic reflexes, e.g. in encephalomyelitis disseminata, stroke or other pathologies of the central nervous system. A long-term neuronal overstimulation leads to a fascial remodeling, increase of collagen fibers, and more rigid and less hydrated connective fascial tissue. This fits with altered tissue microperfusion, hydration and nutrition (Chapter 9). From a purpose- or function-oriented teleological point of view, fascial hypertrophy is not only a compensation for muscle overstimulation, but also a reaction to inflammatory processes. In other words, the strategy of our bodies is based on the principle that fast healing is more important than function. This is not only visible in muscle fascia, but also in the scarring of skin lesions, or in organs, such as connective tissue replacing heart muscle after myocardial infarction, leading to a persistent loss of cardiac output. Hence, fascial hypertrophy is embedded in the survival strategy of our bodies.

A complete inhibition of innervation can be observed during general anesthesia in the hospital. Here, neuromuscular blocking agents are used for induction and tracheal intubation. For training purposes, it is helpful for any movement professional to visit an operating theatre to gain an impression of the effect of a complete neuromuscular block on myofascial tissue.

Surprising facts about fascial physiology and biochemistry

Short-term modulators

Several factors affect the function of fascia, some of which are detailed below.

Physical strain

Histological examination of fascial tissue samples shows that fascia is mainly composed of collagenous fibers embedded in a matrix. Fascial tissue also contains cellular components, such as nerves, blood vessels, fat cells, fibroblasts and myofibroblasts. These cells have contractile properties and are involved in tissue remodeling and reaction to physical strain. Fibroblasts can express contractile proteins under special conditions such as mechanical or chemical stimuli (Figure 2.2). Several growth factors are expressed in response to mechanical loading. Examples are transforming growth factor- β 1 (TGF- β 1), insulin-like growth factor-I (IGF-I) and connective tissue growth factor (CTGF). These factors induce the proliferation of fibroblasts, myofibroblasts and

the synthesis of collagenous fibers (Table 2.1). These processes are similar to those found in skin wound healing (Klingler et al., 2012).

However, there is experimental evidence that repetitive strain leads to changing biomechanical properties independent of cellular contractility. The research shows that the variations observed in peak force and stiffness in repetitive strain originate from changes in tissue hydration due to loading and unloading of the fascial tissue. Hence, cellular aspects and temporarily changed matrix hydration play a role in the biomechanical properties of fascial tissue. The clinical implications are considered as enhanced load transfer promoting a more direct muscle–tendon interaction (Schleip et al., 2012).

The use of stretching in sports science causes debate. For fascial tissue, it can be assumed that stretching immediately before exercise increases the range of motion, possibly by influencing tissue hydration and fiber alignment.

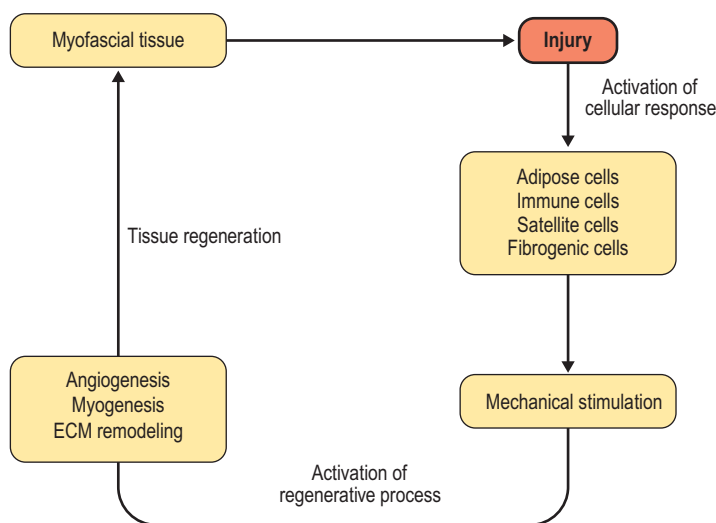


Figure 2.2 Regeneration of myofascial tissue

Following injury, a plethora of processes are activated to regenerate the lost myofascial tissue and restore its original biochemical features: immune cells are recruited to the site of injury and stimulate satellite cells, angiogenesis and remodeling of extracellular matrix (ECM); fibrogenic cells (fibroblasts and myofibroblasts) are activated and remodel ECM; satellite cells exit the quiescent state, proliferate and differentiate into mature muscle fibers; and adipose cells remove cell debris and stimulate satellite cells. In this context, mechanical stimulation promotes myogenesis, angiogenesis and ECM remodeling.

Chapter 2

Table 2.1 Factors influencing the fascial system

Factors influencing the fascial system	Example	Clinical impact
upper motoneuron	injury in stroke	spastic palsy
lower motoneuron	injury in herniated lumbar disc	flaccid palsy
extrapyramidal system	Parkinson syndrome	rigid, hypokinetic movement disorder
cerebral hormones	serotonin, dopamine	mood, need for movement
microenvironmental conditions	pH, temperature, hydration	effects on extracellular matrix and (myo)fibroblasts
humoral factors	transforming growth factor- β 1 (TGF- β 1), connective tissue growth factor (CTGF), relaxin, interleukins, cortisol	effects on extracellular matrix and (myo)fibroblasts
physical strain	sporting activity, stretching	effects on all discussed levels

This might be helpful in sports such as swimming (e.g. front crawl). Here, optimum propulsion significantly depends on the leverage produced by elongation of the shoulder and arm. Other sports, such as hurdling, may benefit from firm fascial structures because of more direct force transmission and reduction of ground contact time (Chapter 10).

Temperature

Temperature is highly variable in myofascial tissue and is unevenly distributed throughout the body. Moreover, myofascial tonus strongly depends on temperature. The core temperature is regulated between 36.5 and 38.5°C. Outside the body core, temperature greatly depends on environmental conditions and muscle activity. At rest, a limb can cool down to roughly 30°C, whereas during sports 40°C can be reached. Higher temperatures, for example in fever or sustained exercise, increase metabolism by roughly 10–20% per degree Celsius. Contraction in skeletal muscle is solely dependent on calcium release from internal stores, the sarcoplasmic reticulum. Calcium unmask the binding site of the contractile enzyme myosin, which is a temperature-dependent ATPase. When adenosine-triphosphate (ATP) is consumed by myosin,

heat is produced and the muscle fibers contract. ATP is replenished by cellular respiration and glycolysis. In turn, these biochemical reactions lead to accumulation of the metabolite lactate, which promotes the synthesis of growth factors and collagen (see below).

High temperatures lead to enhanced excitability and calcium turnover in muscle cells. On excised myofibers, an increased basal tension as well as increased twitch force can be observed. Interestingly, the fascial components show a different reaction to temperature changes. Fascia shows a higher peak force and stiffness in cold conditions. In other words, an important fascial feature is heat relaxation.

At first glance, the different temperature characteristics of muscle fibers and fascial tissue appear to be conflicting. However, the opposite is true. Under resting, i.e. cool conditions, the viscoelastic properties of fascial tissue are adapted to serve stabilization and load-bearing function (Figure 2.3). When myofascial tissue is warmed up, i.e. during sport or activity, fascia demonstrates heat relaxation, allowing an expansion of movement range. Relaxation of fascial tissue, induced by low temperature, can promote injuries in some instances, probably because of a

Surprising facts about fascial physiology and biochemistry

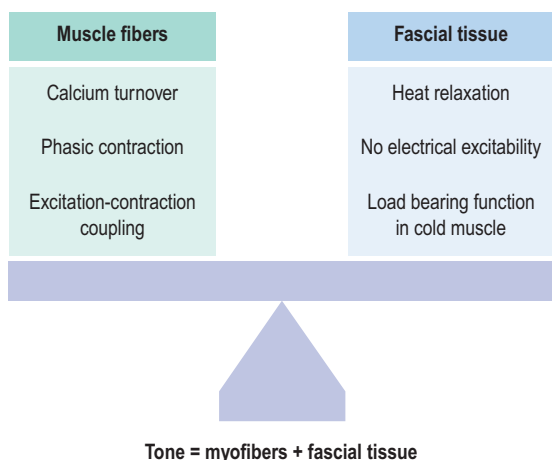


Figure 2.3 Temperature balance

During physical activity, the temperature in myofascial structures outside the body core can increase by up to 10°C. Warmth leads to enhanced skeletal muscle excitability, faster contraction and relaxation parameters, as well as increased force generation. This effect is mainly caused by an increased calcium turnover and temperature-induced facilitation of enzymatic processes, promoting contractile activity. In fascial tissue, however, higher temperature leads to heat relaxation and reduced myofascial stiffness *in vitro*. Given that there is no voluntary innervation, this effect can also be observed *in vivo*. The lesson we learn from the differential temperature effects on fasciae and myofibrils also helps us to understand passive muscle tone. At rest, i.e. in cold conditions, there is an augmentation of the load-bearing function of fascial tissue, whereas in sports, temperatures of greater than 40°C are reached, allowing fascial release and increase in the range of motion.

more rigid tissue response (Wearing et al., 2014). On the other hand, this strongly depends on the degree and type of tissue stress.

Tissue pH

pH is an indicator of acidic or alkaline conditions. It is tightly regulated in all biological systems because it is a key element of all organic processes in the body, such as oxygen uptake, coagulation and immune response. A pH-optimum is necessary for effective enzymatic

functions. pH values below 7.35 indicate a high concentration of protons (H^+ ions), i.e. acidic conditions. Blood pH values above 7.45 are defined as alkalosis. Potassium is an ion that can penetrate the lipid bilayer of cell membranes and, therefore, serve as a counter ion to protons. In other words, potassium concentration increases with dropping pH.

Physical exercise leads to a drop in myofascial tissue pH. This phenomenon is due to the production of lactate and carbon dioxide by glycolysis and cellular respiration. Acidic metabolites accumulate and are exhaled by the lungs as carbonic acid and are more slowly excreted by the kidneys. To a lesser extent, acids are egested by the skin, liver and bowels. Deeper breathing after exercise is not only necessary for replenishment of oxygen but also for elimination of carbonic acid. Abnormal chronic breathing patterns, which influence pH regulation, may also have an impact on myofascial tissue (Chaitow, 2007). Muscle activation can lead to tissue acidosis, as can inflammations of various causes, whether they are traumatic, infectious or autoimmune.

This theoretical background is necessary to understand, for example, what a runner feels when moving at different levels of speed. In the first few seconds after starting a workout there is sufficient ATP available to cover the demand for energy. Approximately half a minute later, the muscular energy buffers myoglobine and creatine phosphate (CP) are activated, and they are used up until the respiratory chain and glycolysis are fully turned during the next few minutes of exercise. In other words, a 100 m runner does not depend on oxidative metabolism during their run. Of course, the energy buffers are replenished after the 100 m run, and each athlete uses oxygen for this. By contrast, a marathon runner depends heavily on their oxidative metabolism because it is a long-lasting exercise.

Chapter 2

If the energy demand exceeds the oxidative metabolism, the body will use up the aforementioned buffers and switch to anaerobic glycolysis. There is a build-up of acidic metabolites, most notably lactate. Acidic metabolites physiologically have negative feedback on muscle excitation-contraction coupling and also mediate fatigue.

Lactate

Lactate plays a major role in tissue regeneration, not only because it promotes blood perfusion and exchange of nutrients, but also because of its effects upon collagen synthesis and angiogenesis.

In vitro data show that acidic conditions enhance contractility of myofibroblasts (Pipelzadeh and Naylor, 1998). This feature is important in tissue repair. The opposite effects can be observed in skeletal muscle fibers. Here, increasing concentrations of protons, potassium and lactate reduce contractility and mediate fatigue by inhibition of membrane excitability and the enzymatic activity of the myosin ATPase, as well as reduction of the glycolytic rate and calcium turnover (Gladden, 2004).

Lactate concentration is increased nearly 2-fold in the healing of Achilles tendon rupture, indicating a key factor of the metabolic response after tissue damage. Lactate has been shown to stimulate vascular endothelial growth factor (VEGF) production. Furthermore, it induces myofibroblast differentiation via pH-dependent activation of transforming growth factor- β (TGF- β), as shown in lung fibrosis. The authors, therefore, speculate that the acidotic microenvironment and the metabolite lactic acid play an important role in connective tissue remodeling (Trabold et al., 2003).

Lactate is an agonist on free nerve endings, which are numerous in fascial tissue. Hence, the “muscle soar” during and shortly after exercise is mainly due to a lactate mediated stimulation of fascial nociceptors.

Long-term modulators

Extracellular matrix

The extracellular matrix (ECM) is a dynamic complex that constantly modifies its viscoelastic properties. It adapts to changes in physiological as well as mechanical demands. It is composed of collagen fibers and a gelatinous ground substance made up of glycoproteins and proteoglycans. The latter serves as mechanical buffer system similar to a water bed. Hydration can influence the mechanical properties of the ECM (Figure 2.4). TGF- β promotes the build-up of ground substance as well as regulating the expression of catabolic enzymes and other mediators.

From a mechanical perspective, it can be observed that, shortly after strain in the Achilles and patella tendons, the tendon's cross-sectional area decreases significantly. In transverse strain, recovery is prolonged. The authors speculate that the decrease in tendon diameter resembles a squeeze of water. The resulting rehydration in recovery may be an important contributor to a slow-acting exchange of nutrients, electrolytes and other humoral factors, such as cytokines (Wearing et al., 2014).

Notably, the excretion of humoral factors depends on the direction of strain on (myo)fibroblasts. Irregular strain, such as in injury, leads to a significantly higher release of interleukin-6, a substance involved in regenerative processes and an increased production of nitric oxide (NO), which is a gaseous neurotransmitter and vasodilator (Murrell, 2007).

Surprising facts about fascial physiology and biochemistry



Figure 2.4 Physiology and biochemistry of myofascial tissue

Fascial tissue forms an interconnected 3-dimensional network throughout the whole body. Physiological and biochemical processes determine tissue characteristics such as collagen turnover, extracellular matrix hydration and stiffness of fascial components. The photograph shows a fictitious illustration of fascia on the lower leg. Here, a loose "fascial stocking" might contribute to the development of varicosis. On the other hand, tight fascial tissue can lead to exercise-induced pain, mismatch of blood perfusion and muscle metabolism in the calf. Indeed, compartment syndrome is a feared post-exercise complication in running athletes.

Growth factors

In fascial tissue, several growth factors are produced in response to mechanostimulation, for example, in sports, but also after tissue damage. The interaction of multiple humoral factors is a complex orchestra of biochemical processes, which does not allow simple cause-and-effect conclusions. The most prominent growth hormones in fascial tissue are TGF- β , IGF-I,

platelet-derived growth factor and CTGF. Growth factors regulate, in collaboration with several cytokines (e.g. interleukin 1, 6 and 8), the proliferation and differentiation of fibroblasts and myofibroblasts, as well as the production of collagen and extracellular matrix proteins. These humoral factors also influence blood perfusion, tissue hydration, pain generation and nutrition, as well as cell migration due to chemotactic properties (Kjaer et al., 2009).

Chapter 2

Hormones

Hormonal influences, as well as growth factors, are important to fascial tissue. Insulin has anabolic effects and enhances (myo)fibroblast proliferation *in vitro*. Estrogen and thyroid hormone receptors have also been found in fibroblasts. Preliminary experiments suggest that thyroid hormones promote fibroblast expansion and counteract apoptosis in connective tissue. Estrogen challenge leads to a greater than 40% reduction in collagen synthesis, a reduction in fibroblast proliferation *in vitro*, and a reduction of growth factor bioavailability. Female subjects on oral contraceptives had a reduced collagen synthesis in response to sporting activity. The authors speculate that this finding might explain the greater risk to female athletes of certain types of injury, for example, cruciate ligament rupture and pelvic joint instability (Hansen et al., 2009).

Endogenous cortisol is the main opponent of insulin and a regulator of tissue metabolism and inflammation. Several studies show that high doses of corticosteroid hormones and derivatives lead to a reduction in the proliferation and activity of fibroblasts and collagen synthesis, leading to delayed and hampered healing response after tissue damage. Conversely, cortisol at physiological levels is indispensable for cellular energy metabolism. Hence, the dose-dependent interaction with cells and other factors is decisive for the anabolic or catabolic effects of corticosteroid hormones. Corticosteroids can be an effective treatment for the prevention of pulmonary fibrosis, for example, in asthma. However, because of the catabolic effects at high concentrations, injections of corticosteroids in injuries or chronic pain in fascial structures should be avoided whenever possible.

Relaxin is a dimeric peptide hormone that is structurally related to the insulin family of peptides, and which is upregulated during pregnancy, for

example. The most consistent biological activities of relaxin are a regulatory effect on collagen synthesis and the stimulation of collagen breakdown. Interestingly, genetically engineered mice which do not produce relaxin at all are characterized by a significant increase in type I and III collagen, resulting in severe scleroderma and interstitial fibrosis, leading to multi-organ failure. This effect could be inhibited by administration of recombinant relaxin (Samuel et al., 2005).

Fibroblasts of the fascia and connective tissue also express cannabinoid receptors (CBRs), and enzymes metabolizing endocannabinoids. The upregulation of the endocannabinoid system in fascial tissue reduces inflammation, fibrosis and pain. This evidence further supports the use and the development of CBR-targeted pharmaceutical compounds for the treatment of fascial diseases (Fede et al., 2016).

Regeneration of muscle and fascia

Intense physical exercise and normal daily activities can be harmful to myofascial structures. Indeed, sportsmen and women often experience muscle strain (Chapters 16 and 17). Once damaged, skeletal muscle and fascial tissues can regenerate. This process is achieved by the synchronized activity of many different elements, such as muscle cells (myofibers and muscle satellite cells), fascial cells (mainly fibroblasts and myofibroblasts), immune cells (mainly polymorphonuclear leukocytes and monocytes/macrophages) and components of the extracellular matrix (mainly protein fibers and growth factors) (Figure 2.2) (Schiaffino and Partridge, 2008).

The ECM in the human body does not have a permanent structure and composition. On the contrary, it varies regularly, depending on environmental and biochemical stimuli. In particular, ECM undergoes repeated cycles of

Surprising facts about fascial physiology and biochemistry

breakdown and assembling. ECM degradation is due, mostly, to the activity of the metalloproteinases and a disintegrin and metalloprotease released extracellularly by different types of cells, such as fibroblasts, immune cells, and skeletal muscle cells. On the other hand, ECM production relies mainly on the secretion and assembling of fibrous proteins and macromolecules by fibroblasts and myofibroblasts. As a result, the ECM network can be restructured at any time to best fit the body's needs, and regenerated after injury (Schiaffino and Partridge, 2008).

After myofascial tissue injury, fibroblasts are recruited to the lesion site where they proliferate and build up ECM, which confers mechanical resistance to the damaged area, and secrete growth factors, which stimulate the proliferation and differentiation of myogenic cells into mature muscle fibers. In addition to the structural role of ECM, it also represents a guide and molecular switch for cells. Moreover, ECM fragments can act as chemotactic agents for the migration of cells into the injury site.

The regeneration of skeletal muscle tissue mainly depends on satellite cells (SCs), the stem cells of skeletal muscle tissue. SCs reside in a quiescent state between the sarcolemma of myofibers and the basal lamina, a fibrous layer of ECM proteins surrounding muscle fibers. As a consequence of muscle injury, SCs are activated, proliferate and differentiate into mature muscle fibers, thus replacing the lost muscle tissue.

Inflammation is a very important process in the regenerative activity of skeletal muscle and fascial tissues. Indeed, at the site of injury, immune cells such as leukocytes, monocytes/macrophages and T cells are recruited and cytokines are released. In turn, the immune response activates muscle satellite cells, stimulates the formation of new myofibers, induces angiogenesis, promotes ECM remodeling and inhibits fibrosis. Therefore,

a well-orchestrated immune response is critical for an effective regeneration of myofascial tissue, whereas a dysregulated immune reaction results in the accumulation of fibrotic tissue (Zügel et al., 2018).

There is a reciprocal influence of skeletal muscle and fascial tissues in regenerative processes of myofascia. This is due to a strict structural and functional interaction between the two tissues. According to a recent hypothesis, the regenerative potential of skeletal muscle is also influenced by adipose and nerve cells (Zullo et al., 2017).

Interestingly, adipose cells, in the form of fibro/adipogenic progenitors and adipocytes, are present in skeletal muscle tissue, where they remove cell debris and stimulate satellite cells after tissue damage and release adipokines, which have an important effect on skeletal muscle regeneration (Joe et al., 2010).

Traumatic events in myofascial tissue can also affect nerves. In this scenario, ECM guides the damaged nerves in the formation of new neuromuscular junction. In turn, growing axons modulate gene expression in the myotubes involved in the regeneration process. In the fascial and skeletal muscle tissues, protein fibers and cells retain a well-defined spatial orientation. The appropriate alignment of these elements is essential for the biomechanical properties of these tissues. During the regeneration of myofascia, mechanical force plays an important role in the regeneration process (Bove et al., 2016; Klingler et al., 2012).

Indeed, mechanical stress, in the form of muscle loading and passive therapies such as stretching, massage and manipulation, strongly influences the activity of muscle cells and fibrogenic cells, the formation of ECM, and the orientation of ECM's fibers and cells. Therefore,

Chapter 2

the application of mechanical stimuli to the myofascial tissue affects its biomechanical features and enhances the regeneration process following injury (Figure 2.2) (Zügel et al., 2018).

Regeneration of myofascial tissue is a very sensitive process. It is compromised in the presence of counteracting factors, such as an unfavorable physical or biochemical environment, or a disease, thus leading to formation of non-functional fibrotic tissue or to fibro-adipogenic degeneration (Klingler et al., 2012). This alteration of tissue structure and composition leads to a dysfunctional myofascia, with reduced performances, decreased mechanical resistance, and increased risk of chronic pain development (Zullo et al., 2017).

Clinical summary

- Sensory input from myofascial tissue is indispensable for nervous control of movement.

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- Myofascial tone depends on neuronal excitation, activation level of intracellular myoplasmic calcium turnover, mechanical properties of myofibers and, to a great extent, on the amount and nature of interwoven, adjacent and attached fascial components.
- Sporting activity leads to significant micro-environmental changes, such as temperature increase, accumulation of metabolites, hydration shift in the extracellular matrix and alteration of protein expression including growth factors and cytokines. Adaptation processes in fascial tissue can explain parts of post-exercise phenomena such as muscle soreness.
- Knowledge about fascial tissue may help to improve movement and should be integrated in sport and exercise science and practical application. Depending on the type of sport, the fascial characteristics may be decisive for optimum performance.

Surprising facts about fascial physiology and biochemistry

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